

HOMEWORK - 03 & 04

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Q1.

Given a table containing information about bank deposits and withdrawals made using Paypal, write a query to retrieve the final account balance for each account, taking into account all the transactions recorded in the table with the assumption that there are no missing transactions.

transactions Table:

Column Name	Type
transaction_id	integer
account_id	integer
amount	decimal
transaction_type	varchar

transactions Example Input:

transaction_id	account_id	amount	transaction_type
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123	101	10.00	Deposit
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124	101	20.00	Deposit
-----	-----	-------	---------

125	101	5.00	Withdrawal
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126	201	20.00	Deposit
-----	-----	-------	---------

128	201	10.00	Withdrawal
-----	-----	-------	------------

128	201	10.00	Withdrawal
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Example Output:

account_id	final_balance
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101	25.00
-----	-------

201	10.00
-----	-------

201	10.00
-----	-------

SOLUTION:

```

SELECT ACCOUNT_ID,
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File display
CASE
WHEN TRANSACTION_TYPE=Deposit' THEN AMOUNT
WHEN TRANSACTION TYPE=Withdrawal' THEN -AMOUNT
ELSE 0
END
) AS FINAL_BALANCE
FROM TRANSACTIONS
GROUP BY ACCOUNT ID

```

Q2.

A phone call is considered an international call when the person calling is in a different country than the person receiving the call.

Write a query to find the percentage of phone calls that are international.
Round the result to 1 decimal place.

Assumption:

- The caller_id in the phone_info table refers to both the caller and the receiver.

phone_calls Table:

Column Name	Type
caller_id	integer
receiver_id	integer
call_time	timestamp

phone_calls Example Input:

caller_id	receiver_id	call_time
-----	-----	-----
1	2	2022-07-04 10:13:49
1	5	2022-08-21 23:54:56
5	1	2022-05-13 17:24:06
5	6	2022-03-18 12:11:49

phone_info Table:

Column Name	Type
-----	-----
caller_id	integer
country_id	string
network	string
phone_number	string

phone_info Example Input:

caller_id	country_id	network	phone_number
-----	-----	-----	-----
1	US	Verizon	+1-212-897-1964
2	US	Verizon	+1-703-346-9529
3	US	Verizon	+1-650-828-4774
4	US	Verizon	+1-415-224-6663
5	IN	Vodafone	+91-7503-907302
6	IN	Vodafone	+91-2287-664895

Example Output:

| international_calls_pct |

|-----|

| 50.0 |

SOLUTION:

```
SELECT
ROUND(100.0*COUNT (
CASE WHEN C.COUNTRY_IDOR.COUNTRY_ID THEN 1
END
) / COUNT(*), 1)
AS INTERNATIONAL_CALLS_PCT
FROM PHONE_CALLS AS P
JOIN PHONE_INFO AS C
ON P.CALLER_ID=C.CALLER_IDD
JOIN PHONE_INFO AS R
ON P.RECEIVER_ID=R.CALLER _ID
```